Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ with Prime Factor Blocks

Polynomial Equations & Factoring • Standard: CCSS.MATH.HSA.APR.A.1

---

1. MATHEMATICAL CONCEPT & STACK APPROACH: Factor the constant term with Prime Factor blocks, sketch the $(x + p)(x + q)$ area grid, and find the roots of the quadratic equation.

A polynomial $ax^2 + bx + c$ mirrors base-10 place value with variable powers (x^2, x^1, x^0) in place of 10^2, 10^1, 10^0. Factoring $x^2 + 5x + 6$ into $(x + 2)(x + 3)$ decomposes the constant term into prime factors ($2 \times 3 = 6$) whose sum equals the linear coefficient ($2 + 3 = 5$).

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Prime Factor Studio (PRIME-01) & Algebraic Area Diagram

Hands-On Setup: Stack Prime Factor blocks [2] and [3] ($2 \times 3 = 6$) in the comparator lid and sketch the 2×2 area grid for $(x + 2)(x + 3) = x^2 + 3x + 2x + 6$.

Student Prediction (Before Calculator): Prime factors of 6 are 2 and 3, and their sum is $2 + 3 = 5$. The factored form is $(x + 2)(x + 3)$ with roots $x = -2$ and $x = -3$.

3. CALCULATOR EXECUTION (SC-32 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: 1 ENTER 5 ENTER 6 g POLY

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
1	1	Coefficient a = 1 entered into X
ENTER	1	1 lifted to Y
5	5	Coefficient b = 5 entered into X
ENTER	5	b in Y, a in Z
6	6	Coefficient c = 6 entered into X
g POLY	-2	Press [g:] [POLY]: Root 1 = -2 in Y (factor: x + 2), Root 2 = -3 in X (factor: x + 3)!

4. MATHEMATICAL CONTEXT & REAL-WORLD MOTIVATION (Al-Khwarizmi's Geometric Squares & The 1535 Public Algebra Duel in Milan)

In 820 AD in Baghdad, Al-Khwarizmi solved $x^2 + 10x = 39$ by drawing literal geometric squares and rectangles ('completing the square')—and in 1535 in Milan, Niccolò Tartaglia defeated Antonio Fiore in a 30-problem public algebra duel overnight! By mapping polynomial coefficients onto the Place Value Tray and factoring constant terms with Prime Factor blocks, students see why $(x + p)(x + q)$ works geometrically.

Core Investigation (Student Warm-Up A):

Factor $x^2 + 7x + 12$ by decomposing 12 with Prime Factor blocks to find the factor pair whose sum is 7, and find both roots of $x^2 + 7x + 12 = 0$.

Signed Factor Pairs: Factoring Trinomials with Negative Terms

Factoring Quadratic Trinomials • Standard: CCSS.MATH.HSA.REI.B.4

1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: When a quadratic has a negative constant or linear term (such as $x^2 - x - 6$), the factor pair must have opposite signs (-3 and +2) so their product is -6 and their sum is -1. You can check both roots on the calculator after factoring on paper.

Blocks & Prediction: Set up coefficients for $x^2 - x - 6$ on the Place Value Tray and factor $|c| = 6$ into Prime Factor blocks [2] and [3]. Prediction: Assign signs to the factor pair (2, -3) so their product is -6 and their sum is $2 + (-3) = -1$: factors are $(x - 3)(x + 2)$.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[1] [ENTER] [1] [+/−] [ENTER] [6] [+/−] [↗] [POLY]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[1]	1	Coefficient a = 1
[ENTER]	1	1 lifted to Y
[1]	1	Enter magnitude of b
[+/−]	-1	Coefficient b = -1 in X
[ENTER]	-1	b in Y, a in Z
[6]	6	Enter magnitude of c
[+/−]	-6	Coefficient c = -6 in X
[↗] [POLY]	3	Press [g:] [POLY]: Root 1 = 3 in Y (factor: x - 3), Root 2 = -2 in X (factor: x + 2)!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Expanding $(x + 2)(x + 3)$ as $x^2 + 6$ by forgetting the two linear cross-terms (2x and 3x).

Hands-On Block Remediation: Place coefficients [1], [5], [6] in the x^2, x^1, x^0 sockets of the Place Value Tray and factor 6 into Prime Factor blocks [2] and [3] so students see that $b = 2 + 3 = 5$ and $c = 2 \times 3 = 6$.

Misconception 2: Using the subtraction operator [−] instead of the change-sign key [+/−] when entering negative polynomial coefficients into [POLY].

Hands-On Block Remediation: Point out that [−] immediately subtracts X from Y, whereas [+/−] negates the number currently in Register X without collapsing the stack.

4. EXTENSION MATHEMATICAL INSIGHT

When a quadratic has a negative constant or linear term (such as $x^2 - x - 6$), the factor pair must have opposite signs (-3 and +2) so their product is -6 and their sum is -1. You can check both roots on the calculator after factoring on paper.

Algebraic Expressions, Base-x Polynomials & Quadratic Factoring

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: QUADRATIC AREA MODEL & PRIME FACTORING RULES

- A polynomial $ax^2 + bx + c$ mirrors base-10 place value with variable powers (x^2, x^1, x^0) in place of 10^2, 10^1, 10^0.
- To factor $x^2 + bx + c$ into $(x + p)(x + q)$, use Prime Factor blocks to factor c into two numbers p and q whose product is c and whose sum is b.
- When c is negative, p and q have opposite signs; when c is positive and b is negative, both p and q are negative.

Key Sequence: `1` `ENTER` `5` `ENTER` `6` `↷` `POLY`

Investigation 1: Factor $x^2 + 6x + 8$ by decomposing 8 into a factor pair that sums to 6, and find the roots.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 2: Factor x^2 - 4 (difference of two squares, where the linear coefficient b = 0).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 3: Factor $x^2 + 5x + 4$ into two binomials and state the solutions to $x^2 + 5x + 4 = 0$.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 4: Factor x^2 - 5x + 6 using signed factor pairs and state both roots.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Unit 08: Algebraic Expressions, Base-x Polynomials & Quadratic Factoring — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 7–8

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): Factor $x^2 + 7x + 12$ by decomposing 12 with Prime Factor blocks to find the factor pair whose sum is 7, and find both roots of $x^2 + 7x + 12 = 0$.

Core Solution: $3 \times 4 = 12$ and $3 + 4 = 7$; factored form $(x + 3)(x + 4)$, roots $x = -3$ and $x = -4$.

Core Check Keys: [1] [ENTER] [7] [ENTER] [1] [2] [↷] [POLY]

Extension Challenge (Warm-Up B): Factor $x^2 - 5x + 6$ into two binomials by finding two negative factors of $+6$ whose sum is -5, and state both roots.

Extension Solution: $(-2) \times (-3) = +6$ and $(-2) + (-3) = -5$; factored form $(x - 2)(x - 3)$, roots $x = 2$ and $x = 3$.

Ext. Check Keys: [1] [ENTER] [5] [+/−] [ENTER] [6] [↷] [POLY]

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Factor $x^2 + 6x + 8$ by decomposing 8 into a factor pair that sums to 6, and find the roots.

Expected Answer: $(x + 2)(x + 4)$; roots -2, -4

SC-32 Verification Keys: [1] [ENTER] [6] [ENTER] [8] [↷] [POLY]

Investigation 2: Factor $x^2 - 4$ (difference of two squares, where the linear coefficient $b = 0$).

Expected Answer: $(x - 2)(x + 2)$; roots 2, -2

SC-32 Verification Keys: [1] [ENTER] [0] [ENTER] [4] [+/−] [↷] [POLY]

Investigation 3: Factor $x^2 + 5x + 4$ into two binomials and state the solutions to $x^2 + 5x + 4 = 0$.

Expected Answer: $(x + 1)(x + 4)$; roots -1, -4

SC-32 Verification Keys: [1] [ENTER] [5] [ENTER] [4] [↷] [POLY]

Investigation 4: Factor $x^2 - 5x + 6$ using signed factor pairs and state both roots.

Expected Answer: $(x - 2)(x - 3)$; roots 2, 3

SC-32 Verification Keys: [1] [ENTER] [5] [+/−] [ENTER] [6] [↷] [POLY]

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.